

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-507: Combinatorial Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) In how many ways can four tyres be put on a car? [1]
- (b) A king is placed on the bottom left-hand square of an 8×8 chessboard and is to move to the top right-hand corner square. If it can move only up or to the right, how many possible paths does it have to choose from? [2]
- (c) It is required to seat n people round a table. Show that this can be done in $(n-1)!$ ways. [3]
- (d) A pack of 52 cards is divided among 4 people so that each gets 13 cards. How many such deals are possible? [2]
- (e) If

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

derive from B an 8×8 Hadamard matrix and hence write down the incidence matrix of a $(7, 3, 1)$ -configuration. [3]

- (f) Find the Rook's polynomial of the following board:

$$\begin{array}{cccc} \square & & \square & \text{lacksquare} \\ \square & & \text{lacksquare} & \square \\ \text{lacksquare} & & \square & \square \end{array}$$

where *lacksquare* represents a shaded/unavailable square. [3]

Question 2.

- (a) Suppose that each of k indistinguishable golf balls has to be coloured with any one of n given colors and we have to find the different colourings. Show that the number of different colourings is same as the number of non-negative integral solutions of the equation

$$x_1 + x_2 + \cdots + x_n = k$$

Let $f(n, k)$ be the number of solutions of this equation. Show that:

- (i) $f(n, k) = f(n-1, k) + f(n, k-1)$ for all $n \geq 2, k \geq 2$.
- (ii) $f(n, k)$ is the coefficient of x^k in $(1-x)^{-n}$.

(iii) $f(n, k)$ is the number of ways of choosing $(n - 1)$ places out of $(n + k - 1)$ places.

[10]

(b) In a cricket league of n teams, each team plays each other twice. The number of games played is therefore $2C$, where C is the number of ways of choosing 2 objects from n given objects. Prove that

$$2C = 2[(n - 1) + (n - 2) + \cdots + 2 + 1]$$

and deduce the number of games played in a league of 22 teams.

[4]

Question 3.

(a) By using the identity

$$(1 + x)^{2n} = (1 + x)^n(1 + x)^n$$

and considering the coefficient of x^n on both sides, prove that

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2$$

Verify this in the case $n = 5$.

[7]

(b) Use the binomial theorem to evaluate:

$$(i) \sum_{r=0}^n (-1)^r \binom{n}{r}$$

$$(ii) \sum_{r=0}^n r \binom{n}{r}$$

[7]

Question 4.

(a) Show that, if a graph has $2n$ vertices, each of degree $\geq n$, then the graph has a perfect matching. Show by an example that the converse of this statement may not be true. [6]

(b) State and prove Hall's marriage theorem.

[8]

Question 5.

(a) Let $\{a_n\}$ be a sequence satisfying $a_n = Aa_{n-1} + Ba_{n-2}$, $n \geq 2$, where a_0 and a_1 are given. A and B are constants. If r_1 and r_2 are the roots of $r^2 - Ar - B = 0$, then show that:

$$(i) a_n = k_1 r_1^n + k_2 r_2^n, \text{ if } r_1 \text{ and } r_2 \text{ are real and distinct;}$$

$$(ii) a_n = (k_1 + nk_2)r^n, \text{ if } r_1 = r_2 = r;$$

where k_1, k_2 are constants uniquely determined by the initial conditions.

[7]

(b) Solve the Fibonacci recurrence relation: $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, with $F_0 = 0$, $F_1 = 1$.

[7]

Question 6.

(a) Define derangements. If D_n denotes the number of possible derangements of n symbols, then find D_n .

[7]

- (b) Given a chessboard C , choose any square of C and let D denote the board obtained by deleting from C every square in the same row or column as the chosen square (including the chosen square itself). Let E denote the board obtained from C by deleting only the chosen square. Then show that:

$$R(x, C) = xR(x, D) + R(x, E)$$

[7]

Question 7.

- (a) For a (v, b, r, k, λ) -configuration, show that $vr = bk$, $r(k - 1) = \lambda(v - 1)$, and $b \geq v$. [10]
- (b) Show that:
- (i) A code will detect all sets of h or fewer errors if any two words differ in at least $(h + 1)$ places. [4]
 - (ii) A code will correct all sets of h or fewer errors if any two words differ in at least $(2h + 1)$ places. [4]